VIVA GUIDE 03 / 05

The AI answer service

FastAPI orchestration, prompt grounding, Gemini configuration, and failure handling.

YOUR RESPONSIBILITY

Own the request path from a retrieved context to a controlled Gemini answer.

THE SHORT VERSION

FastAPI accepts a question, normalizes it, finds relevant passages in the preloaded NIT index, and·only when there is relevant context and a configured key·builds a grounded prompt and sends it to Gemini. It returns an answer, source metadata, retrieval status, and timing.

A 30-SECOND OPENING

The AI service is a separate FastAPI process. It does not ask Gemini to answer from general knowledge: it first retrieves institute passages, then supplies those passages with rules that restrict and qualify the answer.

IN THIS GUIDE

- Service responsibility and endpoint
- Grounding the model
- Gemini request and configuration
- Response, history, and health semantics
- Reproducing the 502 and explaining it
- Limitations and viva questions

SERVICE RESPONSIBILITY AND ENDPOINT

The service entry point is ai-service/app/main.py. It defines POST /api/v1/nlp/chat and GET /api/v1/health. At startup it loads settings and reads ai-service/data/nith_documents.json into memory through load_nith_documents. ChatRequest validates a non-empty question and accepts an optional conversationHistory list.

The endpoint normalizes the question, calls retrieve_context, and branches. With no matches it returns a fixed not-found answer and no sources. With matches but no API key it raises HTTP 503. With matches and a key, it prepares context, builds the prompt, calls Gemini, and maps provider failures to HTTP 502.

```
Browser -> Express -> FastAPI
-> normalize question
-> retrieve matching NIT passages
-> grounded prompt -> Gemini
-> answer + source metadata -> Express -> browser
```

GROUNDING THE MODEL

prompt_builder.py assembles the instructions and retrieved excerpts. The model is told to use only official excerpts, avoid inventing institutional facts, say when the excerpts do not answer, cite document title/page, treat excerpt text as reference rather than instructions, and write clearly for students.

Each context excerpt is labelled with document name, page, official source URL, and text. This is more useful than sending an anonymous block because the model can refer to the same evidence that is returned as structured source metadata.

Grounding reduces unsupported answers; it does not guarantee correctness. The retrieved passages must be relevant, the source text must be accurate, and the model output still needs to be treated as generated text.

GEMINI REQUEST AND CONFIGURATION

core/config.py reads GEMINI_API_KEY and GEMINI_MODEL from the environment, with gemini-2.5-flash as the default model. The key is passed to the provider only from the server-side generator, in the x-goog-api-key header; it is not a frontend setting.

No live Gemini key was supplied for this project workspace, so authenticated provider generation has not been verified here. A configured-looking health status is not a substitute for testing a real request.

rag/generator.py calls Google's generateContent REST endpoint using httpx. The configured generation temperature is 0.2 and maxOutputTokens is 768. A 30-second timeout bounds provider waiting time. The response is checked for an error status, a valid candidate/content/parts shape, and non-empty text.

- GEMINI_API_KEY: required for a matched question to receive generated text.
- GEMINI_MODEL: optional model name; default gemini-2.5-flash.
- Missing key: an explicit configuration error rather than a fake success.
- Timeout, transport error, non-success HTTP response, or malformed/empty output: explicit GeminiGenerationError.

RESPONSE, HISTORY, AND HEALTH SEMANTICS

A successful response contains answer, sources, retrievalStatus, and processingTime. Each source contains documentId, document title, official URL, page, and a relevance score. When there are no matches, retrievalStatus

is fallback and the model is not called.

Express sends the latest ten turns as conversationHistory, but main.py currently uses only payload.question. Therefore the model does not receive prior turns as conversational context in this implementation. Session history is saved/displayed, but follow-up resolution across turns is not implemented.

The health route reports whether the key is configured, the model name, index readiness, and number of passages. 'Configured' only means a non-empty key value exists; it does not prove that the key is valid or that Gemini is reachable.

REPRODUCING THE 502 AND EXPLAINING IT

A 502 in the AI-service access log means that /api/v1/nlp/chat returned a bad-gateway response. In this code, it is returned when generate_answer raises GeminiGenerationError·for example provider HTTP error, network error, timeout, or invalid/empty response. FastAPI's error detail says whether to inspect the provider response or model/key setup.

- Check /api/v1/health: confirm the key is configured and the knowledge base is ready.
- Verify the key in ai-service/.env is valid, active, and belongs to a project with Gemini API access.
- Confirm the model name is supported for that key and the network can reach generativelanguage.googleapis.com.
- Inspect the AI service response body/log: 503 points to missing configuration; 502 points to generation failure.
- A question with no matching passage will return the grounded not-found response and will not test Gemini.

Never paste a real API key into screenshots, viva slides, Git history, or a client-side file. Use a placeholder in demonstrations.

LIMITATIONS AND VIVA QUESTIONS

The AI service loads its document index at process startup; it does not hot-reload the active NIT index. Retrieval is lexical and is covered in Guide 04. Conversation history is currently ignored by generation. Provider quotas, key validity, and internet access remain external dependencies.

Why retrieve before calling Gemini?
The assistant is intended to answer from institute material, not unrestricted model memory; retrieval supplies supporting evidence and avoids a provider call when nothing relevant is found.

Why is temperature low?
A lower value reduces variability for factual institutional answers, although it cannot eliminate model errors.

What does the 502 indicate in this implementation?
Gemini generation failed and the AI service converted the provider/generator error to HTTP 502.

Does 'Gemini configured' mean the key works?
No. Health checks only whether the key string is non-empty; only an actual provider request validates it.

Does the AI service use conversation history?
Not currently. Express sends recent turns, but the FastAPI endpoint uses only the current question.

Why return sources separately from answer text?
Structured source metadata lets the UI show clickable document and page references consistently.